

Part 2 & 3 - Forecasting and Evaluation

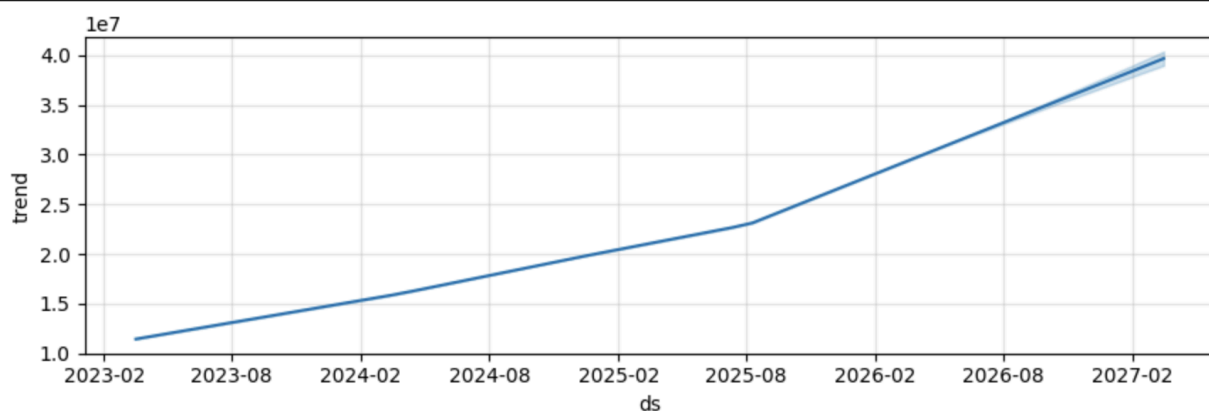
Forecasting Approach

The forecasting task requires revenue predictions for the next 12 months at two granularities: overall and category x market level.

The train period covers March 2023 to March 2025 (731 days).

The test period covers March 2025 to March 2026 (365 days), matching the forecast horizon.

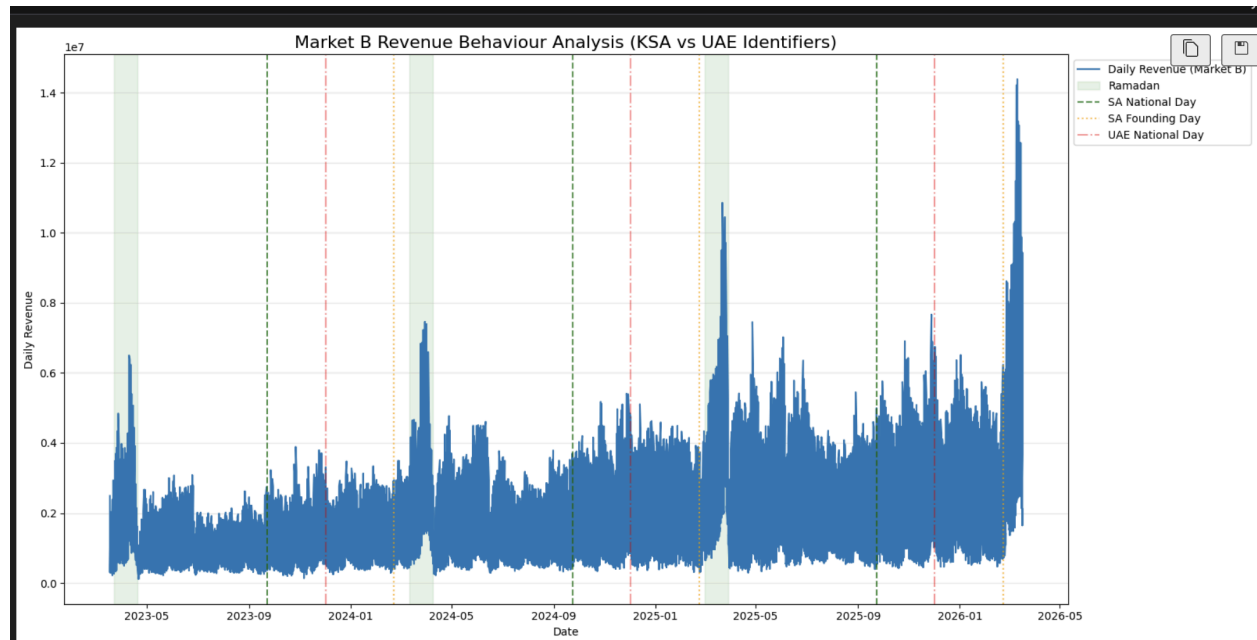
Before selecting a model, two structural properties of this dataset shaped the approach. First, the series exhibits strong linear growth from approximately 12M to 40M in daily revenue over the training period.



Second, demand is driven by structured calendar events specific to the GCC context: Ramadan, Eid al-Fitr, Eid al-Adha, Dubai Shopping Festival, and market-specific national holidays. These

are not random shocks. They are known in advance and recur annually, making them ideal candidates for explicit feature engineering.

Here's a visualization that demonstrates how Ramadan aligns with revenue spikes:



Architecture

Two complementary approaches were used, one per forecast task. For the overall revenue forecast, all 10 category x market series are stacked and predictions summed across combinations per period, giving the model access to the full dataset simultaneously.

For the category x market breakdown, Prophet was run independently per series, so one model per combination, each trained on its own revenue trajectory and calendar regressors.

The two approaches share the same feature engineering pipeline and train/test split.

Primary Model: Prophet

Prophet was selected as the primary model for three reasons. It decomposes trend, seasonality, and holiday effects explicitly, which maps to the structure of this dataset. It supports multiplicative seasonality which is appropriate (for example, revenue growth multiplies each Ramadan on top of the linear trend).

A vanilla baseline was established first with no regressors, then regressors were added to measure the contribution of the GCC calendar features independently.

Overall forecast results:

	MAPE	WMAPE	MAE	Bias
Prophet (vanilla)	17.03%	16.89%	4,131,943	+0.034
Prophet (regressors)	4.39%	4.45%	1,088,790	-0.014

The regressors reduced MAPE by 74% and MAE from 4.1M to 1.09M AED per day. This confirms that the GCC calendar signal is a strong driver of forecast precision and not trend or seasonality shape alone.

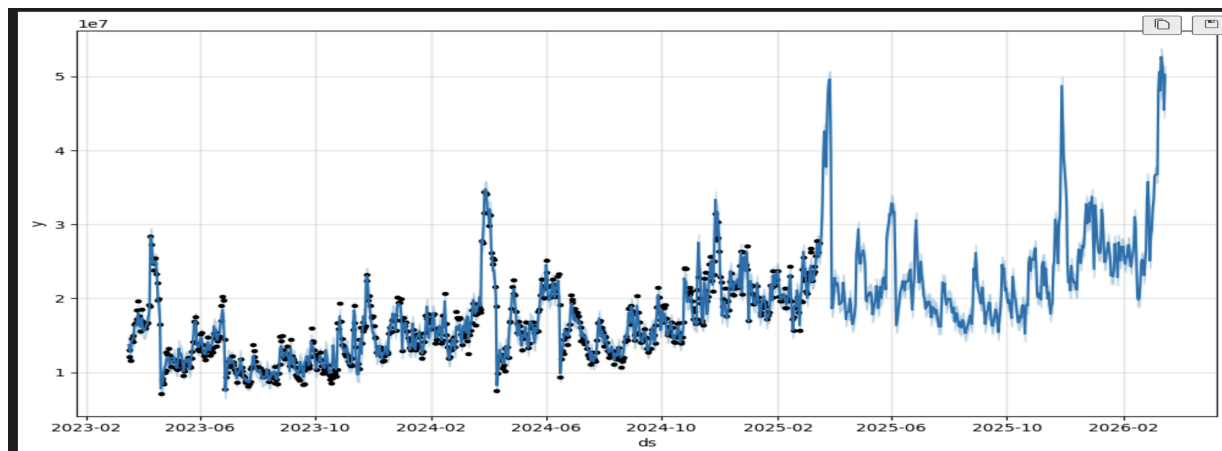
Per-series results with regressors:

Series	MAPE	WMAPE	MAE	Bias
Category_A x Market_A	7.16%	7.19%	451,989	-0.039
Category_A x Market_B	7.68%	7.34%	377,523	+0.058
Category_B x Market_A	9.47%	10.10%	247,679	-0.020
Category_B x Market_B	23.20%	22.62%	355,868	+0.219

Category_C x Market_A	18.61%	18.35%	275,177	+0.117
Category_C x Market_B	11.40%	10.74%	91,672	-0.006
Category_D x Market_A	5.82%	6.01%	139,017	+0.016
Category_D x Market_B	7.65%	7.74%	96,230	+0.028
Category_F x Market_A	8.84%	9.16%	196,453	-0.052
Category_F x Market_B	9.25%	8.87%	84,683	+0.034

High-volume series (Category_A, Category_D) perform well across both markets. The two flagged combinations (Category_B x Market_B, Category_C x Market_A) show high bias driven by sparse signals. These are structurally harder to forecast due to lower volume.

Overall forecast with regressors result:



Alternative Considered: LightGBM

A LightGBM global model was tested as an alternative. All series were stacked vertically with category and market as encoded features alongside the same calendar regressors and lag features at a 365-day anchor.

I hypothesized that patterns emerging across markets and categories might co-vary, therefore benefiting from each other's signals (e.g. How Ramadan impacts a spike in Market A Category B might influence a spike in Market A Category C). This is opposed to an additive model like prophet that cannot learn relationships between series. Each model sees one signal in isolation. Whether we aggregate over daily revenue or build a forecasting model per combination, the model architecture doesn't allow for cross-signal learning.

Overall results:

	MAPE	WMAPE	MAE	Bias
LightGBM	7.92%	9.67%	2,268,999	-0.081

LightGBM underperforms Prophet on this dataset in its current form. The primary issue is hypothesized to be trend capture. Prophet has an explicit trend component. LightGBM sees no direct trend signal: lag_365 tells the model what revenue looked like one year ago but it cannot represent that the business has grown since then. This produces systematic under-forecasting as reflected in the -8.1% bias.

This is a feature engineering gap. The following directions can be explored in a follow-up iteration:

- Explicit trend feature: year-over-year growth rate as a regressor
- Early stopping with a held-out validation set to control overfitting

Prophet remains the primary model given current results. LightGBM can be retained as an active research direction.

Evaluation

Offline evaluation framework

Four metrics are used in combination. Each answers a different question about forecast quality.

MAPE (Mean Absolute Percentage Error) measures average percentage deviation per period. It is scale-free, making it useful for comparing accuracy across series with different revenue magnitudes. It can be unreliable when actuals approach zero, which is why it is used alongside WMAPE.

WMAPE (Weighted Mean Absolute Percentage Error) weights each error by the actual revenue value, so low-revenue periods do not disproportionately inflate the metric. Some combinations might have near zero periods, making this a more robust measure for those series. (This would also better handle a new added market x category).

MAE (Mean Absolute Error) expresses error in AED which is an operationally meaningful number for a planner. A 5% MAPE on a 20M AED day means 1M AED off the OTB budget.

Bias measures the direction and magnitude of systematic error: positive bias indicates consistent over-forecasting, negative bias indicates consistent under-forecasting. For OTB planning, bias is the most consequential metric.

What matters more than accuracy

Bias. In an OTB context, the direction of forecast error has operational consequences.

Systematic over-forecasting leads to excess inventory commitments and capital tied up in stock that does not sell and markdown pressure later in the season.

Systematic under-forecasting leaves purchasing budget on the table, resulting in stockouts on high-demand SKUs during peak periods like Ramadan and Eid.

A model with slightly higher MAPE but near-zero bias is more useful for OTB planning than a low-MAPE model with consistent directional error.

The Prophet model with regressors achieves both: 4.39% MAPE and -1.4% bias on the overall series, indicating the model is well-calibrated with no meaningful directional skew.

Forecasting demand for new products and categories

New products and categories present a cold start problem: there is no revenue history to condition on. Two approaches are applicable depending on the available information.

If a similar existing category is available, its seasonal trajectory serves as a structural prior. This is the fastest path to a usable forecast and is defensible when the proxy category shares meaningful demand drivers with the new one. (e.g. Shoes and Sandals)

If no proxy exists, the position is that forecasting is high-uncertainty until sufficient data is collected.

In this case the deliverable is not a point forecast but a wide prediction interval informed by category-level averages (current mean across supported categories) and domain assumptions.